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LEARNING MANAGEMENT SYSTEM

SOFTWARE DESIGN DESCRIPTION DOCUMENT

VERSION: 1.0

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The following Software Requirements Specification has been accepted and approved by the following:

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Design Description (SDD) document is to describe the architectural and detailed design of the Learning Management System (LMS). This document converts the requirements defined in the Software Requirements Specification (SRS) into a structured system design. It serves as a technical reference for developers, testers, and system evaluators. The SDD explains how system components interact to fulfill functional and non-functional requirements. It provides guidance for implementation and testing activities. This document also supports future system maintenance and enhancements.

1.2 Scope

This document covers the complete design of a web-based Learning Management System intended for academic institutions. The LMS supports user management, course management, content sharing, assignment submission, grading, and announcements. The scope includes system architecture, data design, component design, and user interface design. The system is accessible through standard web browsers using a client-server model. Financial processing and external ERP integration are outside the scope. The design focuses on security, usability, and scalability.

1.3 Overview

This SDD is organized into multiple sections that describe the system design in a structured manner. Section 1 introduces the document and defines its purpose and scope. Section 2 provides a general overview of the system. Section 3 explains the system architecture and decomposition using diagrams. Section 4 describes data design and database structure. Section 5 focuses on component-level design, while Section 6 covers the human interface design. Section 7 provides a requirements traceability matrix.

1.4 Reference Material

This Software Design Description is primarily based on the approved Software Requirements Specification (SRS) for the Learning Management System. UML 2.5.1 standards were referenced for modeling system diagrams. General software engineering principles were followed during design preparation. Web application development guidelines were also considered. Academic documentation standards provided by the Computer Science department were used. No external proprietary documentation was referenced.

1.5 Definitions and Acronyms

LMS stands for Learning Management System and refers to the academic software platform. SDD means Software Design Description. SRS refers to Software Requirements Specification. UI stands for User Interface. RBAC means Role-Based Access Control. UML refers to Unified Modeling Language. DFD stands for Data Flow Diagram.

2. SYSTEM OVERVIEW

The Learning Management System (LMS) is a web-based software application designed to support academic teaching and learning activities. The system provides a centralized platform for managing users, courses, learning materials, assignments, grades, and announcements. It allows administrators to control system configuration and user access. Teachers can manage course content, assignments, and grading processes. Students can access learning resources, submit assignments, and view academic results. The system operates using a client-server architecture and is accessible through standard web browsers. Emphasis is placed on security, usability, and efficient academic workflow management.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The Learning Management System follows a layered client-server architecture to ensure scalability and maintainability. The system is divided into three main layers: Presentation Layer, Application Layer, and Data Layer. The Presentation Layer handles user interaction through a web browser interface. The Application Layer contains business logic such as authentication, course handling, and grading. The Data Layer manages persistent storage using a relational database. Each layer communicates through well-defined interfaces. This separation of concerns improves security and system performance.

3.2 Decomposition Description

The Learning Management System is divided into smaller functional subsystems to reduce complexity and improve clarity. Each subsystem is responsible for a specific academic task. These subsystems work together through the application layer to provide complete LMS functionality. The decomposition follows an object-oriented approach where real-world academic entities are represented as system modules. This design improves code reusability and maintainability. Clear interaction paths between subsystems ensure smooth data flow. UML and DFD diagrams are used to represent system structure and behavior.

Diagram 1: Subsystem Decomposition Diagram (Main Diagram)

Purpose of this Diagram

This diagram shows **how the LMS is divided into smaller subsystems** and how they are connected. It explains **what part of the system does what**.

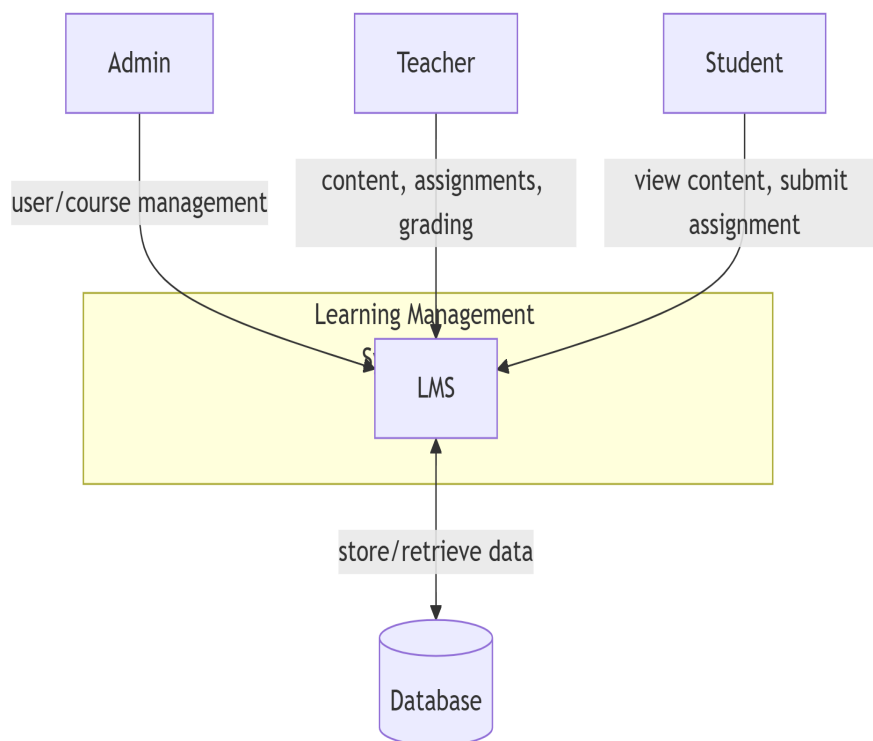


Diagram 2: Level-0 Data Flow Diagram (DFD)

Purpose of this Diagram

This diagram explains **how data flows between users and the LMS**, without internal details.

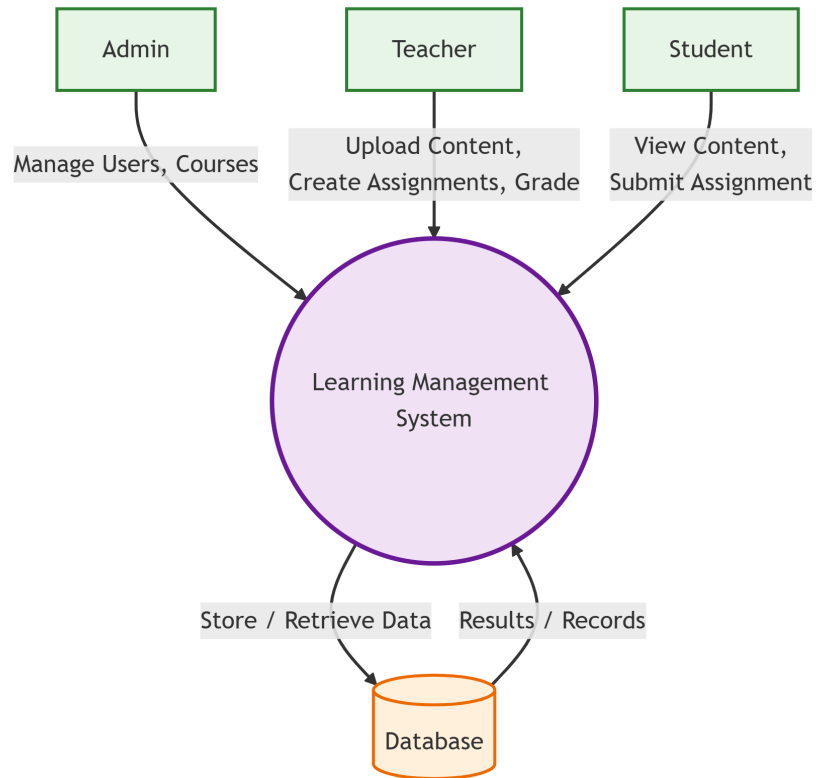


Diagram 3: Level-1 Data Flow Diagram

Purpose

This diagram breaks the LMS into **smaller processes** to show **how each task is handled**.

Level-1 Data Flow Diagram

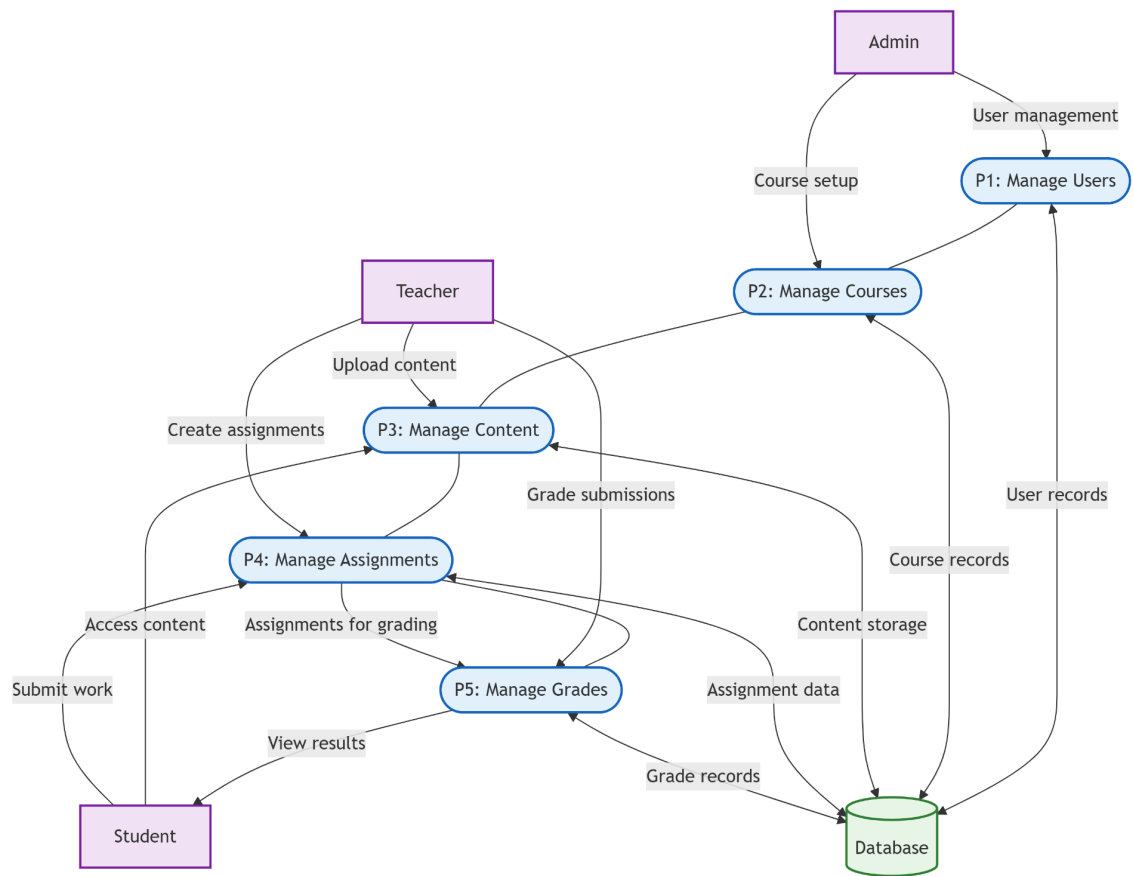


Diagram 4: Class Diagram (Structural Diagram)

Purpose

Shows **system structure using classes and relationships**.

Class Diagram (Structural Diagram)

Class Diagram - Learning Management System

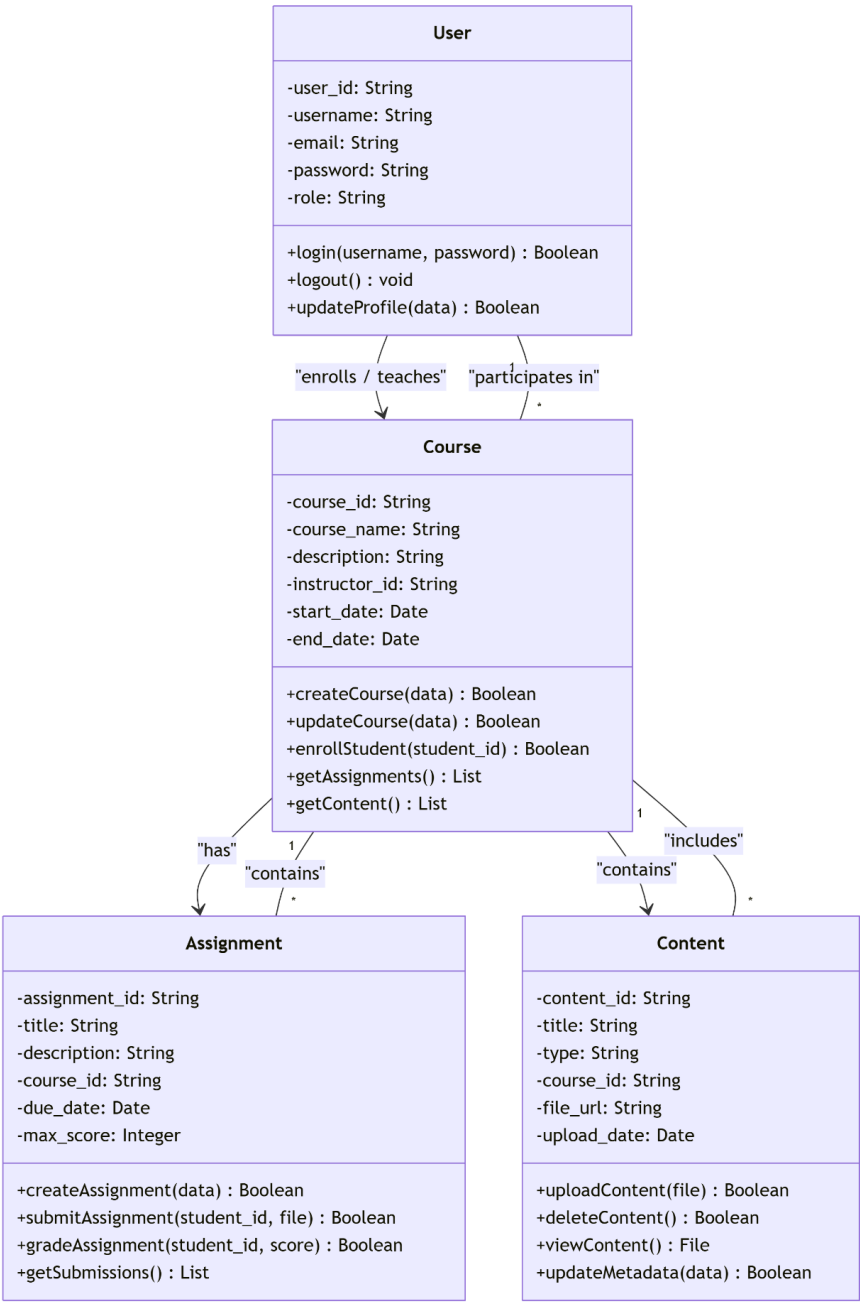
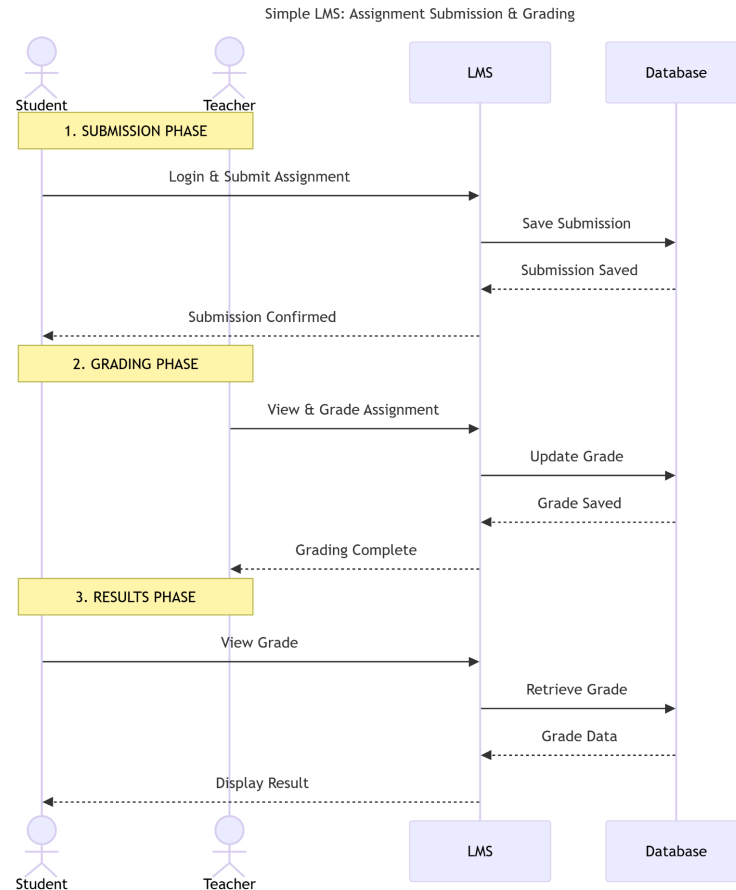


Diagram 5: Sequence Diagram (Assignment Submission)

Purpose

Shows **step-by-step interaction over time**.



3.3 Design Rationale

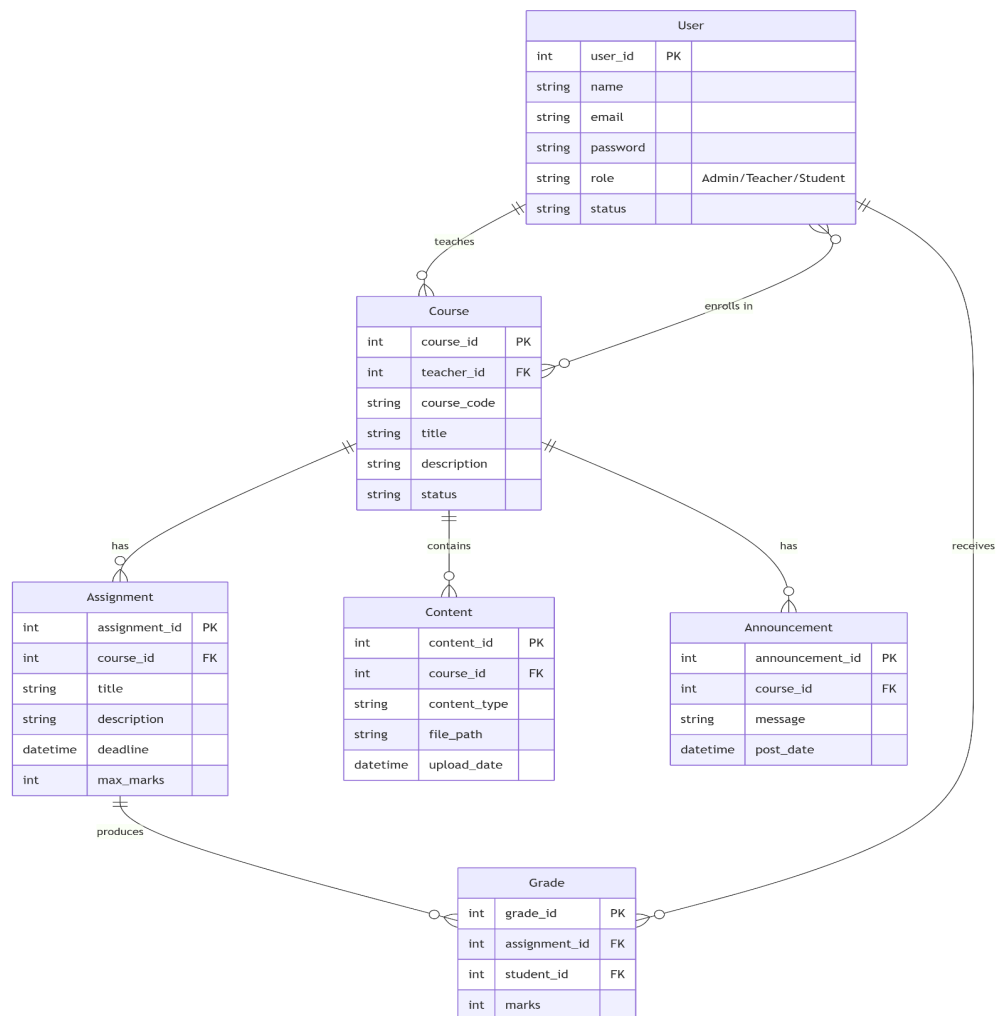
The layered architecture was selected to separate user interface, logic, and data management responsibilities. This design improves maintainability and allows independent updates to system components. Object-oriented decomposition supports reusability and extensibility. Security requirements influenced the use of centralized authentication and role-based access control. Alternative monolithic architectures were rejected due to poor scalability. The selected design balances simplicity and performance. It aligns well with web-based academic systems.

4. DATA DESIGN

4.1 Data Description

The Learning Management System uses a relational data model to store and manage all academic information. System data includes users, courses, assignments, content, grades, and announcements. Each data entity is uniquely identified to avoid duplication and ensure data integrity. Relationships are established using primary and foreign keys. Data is processed through the application layer and stored in the database securely. Only authorized users can access or modify data based on their roles. This design ensures consistency, security, and efficient data retrieval.

ER Diagram



4.2 Data Dictionary

The data dictionary provides a detailed description of all major data entities defined in the Learning Management System as specified in the SRS. It serves as a reference for understanding the structure, attributes, and purpose of system data. The **User** entity stores authentication credentials, roles, and account status. The **Course** entity maintains course information and assigned teacher details. The **Assignment** entity records assignment descriptions, deadlines, and maximum marks. The **Content** entity stores learning materials uploaded by teachers. The **Grade** entity records marks awarded to students. The **Announcement** entity stores course-related notifications. This data dictionary ensures consistency and traceability between the SRS and SDD.

5. COMPONENT DESIGN

This section describes the internal logic of each major system component in a structured manner. Each component is designed as an independent module with clearly defined responsibilities. The components interact through the application layer to deliver complete system functionality. The design follows an object-oriented approach to improve reusability and maintainability. Pseudocode is used to describe component behavior in a simple and understandable way. Local data and decision logic are clearly represented. This design ensures correctness and consistency across system operations.

5.1 User Management Component

Responsibility: Authentication, role assignment, and account management.

Pseudocode: User Login

FUNCTION login(username, password)

 RETRIEVE user FROM database USING username

 IF user EXISTS AND password IS valid

 CREATE session for user

 RETURN access granted

 ELSE

 RETURN invalid credentials message

END FUNCTION

Pseudocode: Manage User

FUNCTION manageUser(userData, action)

 IF action IS create

 STORE userData in database

 ELSE IF action IS update

 MODIFY existing user record

 ELSE IF action IS delete

 REMOVE user record

 RETURN operation status

END FUNCTION

5.2 Course Management Component

Responsibility: Course creation, teacher assignment, and enrollment.

Pseudocode: Create Course

FUNCTION createCourse(courseDetails)

 VALIDATE courseDetails

 STORE courseDetails in database

 ASSIGN teacher to course

 RETURN confirmation

END FUNCTION

Pseudocode: Enroll Student

FUNCTION enrollStudent(studentID, courseID)

 VERIFY student and course existence

 LINK student with course

 UPDATE enrollment data

 RETURN enrollment success

END FUNCTION

5.3 Content Management Component

Responsibility: Uploading and accessing learning materials.

Pseudocode: Upload Content

```
FUNCTION uploadContent(teacherID, courseID, content)
    VERIFY teacher authorization
    VALIDATE content format
    STORE content securely
    SAVE content record in database
    RETURN upload success
END FUNCTION
```

Pseudocode: View Content

```
FUNCTION viewContent(userID, courseID)
    CHECK user enrollment
    RETRIEVE content list
    DISPLAY content to user
END FUNCTION
```

5.4 Assignment Management Component

Responsibility: Assignment creation and submission handling.

Pseudocode: Create Assignment

```
FUNCTION createAssignment(courseID, assignmentData)
    VALIDATE assignmentData
    STORE assignment in database
    SET submission deadline
    RETURN confirmation
END FUNCTION
```

Pseudocode: Submit Assignment

```
FUNCTION submitAssignment(studentID, assignmentID, file)
    CHECK current time against deadline
    STORE submission file
    SAVE submission record
    RETURN submission status
END FUNCTION
```

5.5 Grading Component

Responsibility: Grading assignments and displaying results.

Pseudocode: Grade Assignment

```
FUNCTION gradeAssignment(teacherID, submissionID, marks)
    VERIFY teacher authorization
    VALIDATE marks range
    STORE grades in database
    RETURN grading confirmation
END FUNCTION
```

Pseudocode: View Grades

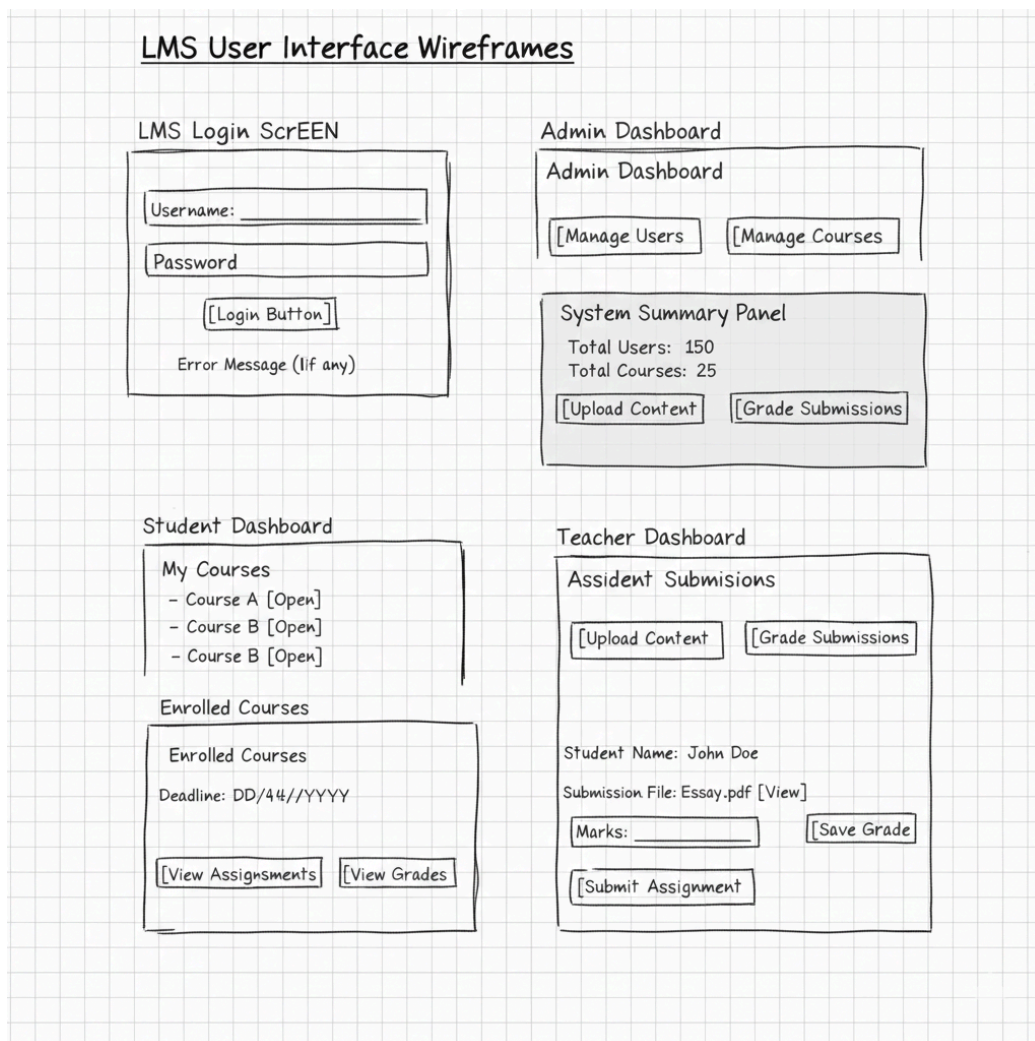
```
FUNCTION viewGrades(studentID)
    RETRIEVE grades from database
    DISPLAY grades to student
END FUNCTION
```

6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The Learning Management System provides a web-based graphical user interface accessible through standard web browsers. The interface supports role-based access for Administrators, Teachers, and Students as specified in the SRS. Users interact with the system using menus, buttons, forms, and dashboards. The interface provides clear feedback messages for successful and unsuccessful actions. Navigation remains consistent across all screens to improve usability. Error messages guide users in correcting mistakes. The design emphasizes simplicity, accessibility, and ease of learning.

6.2 Screen Images



6.3 Screen Objects and Actions

This section describes the user interface objects present on each screen and the actions performed by those objects. Each object directly supports one or more functional requirements defined in the SRS. Actions initiated by users trigger system processes and produce appropriate feedback messages. The interface design ensures that users can complete tasks with minimal steps. Validation and error handling are performed at the screen level where required. All actions are role-based and restricted according to user permissions. This design improves usability and system reliability.

Screen Name	User Role	Screen Objects	User Action	System Response
Login Screen	Admin / Teacher / Student	Username field, Password field, Login button	User enters credentials and clicks Login	System validates credentials and redirects to role-based dashboard or displays error message
Admin Dashboard	Admin	Manage Users button, Manage Courses button	Admin selects an option	System opens selected management module

Teacher Dashboard	Teacher	Course list, Upload Content button, Assignments button, Grade button	Teacher selects course or action	System displays course content, assignment list, or grading interface
Student Dashboard	Student	Enrolled course list, View Assignments button, View Grades button	Student selects course or option	System displays assignments, content, or grades
Assignment Submission Screen	Student	File upload field, Submit button	Student uploads file and submits	System stores submission and shows confirmation message
Grading Screen	Teacher	Submission list, Marks input field, Save Grade button	Teacher enters marks and saves	System stores grades and updates student record

7. REQUIREMENTS MATRIX

This section ensures that every requirement specified in the Software Requirements Specification (SRS) is correctly addressed in the Software Design Document (SDD). The Requirements Traceability Matrix establishes a clear relationship between functional requirements, design components, and system interfaces. It helps verify completeness and consistency throughout the development process. The matrix ensures that no requirement is left unimplemented or unnecessarily duplicated. It also supports impact analysis in case of future requirement changes. Each requirement is traced to specific design sections and diagrams. This traceability improves system reliability and validation.

7.1 Requirements Traceability Matrix Table

SRS Requirement ID	Requirement Description	SDD Section	Design Component / Diagram
FR-01	User authentication and login	Section 5.1, 6.2	User Management Component, Login Screen
FR-02	Role-based access control	Section 5.1, 6.3	User Management Component, Dashboards
FR-03	Course creation and management	Section 5.2, 6.2	Course Management Admin Component, Dashboard
FR-04	Upload and manage learning content	Section 5.3, 6.2	Content Management Teacher Component, Dashboard

FR-05	Assignment creation and submission	Section 5.4, 6.2	Assignment Component, Submission Screen
FR-06	Assignment grading and results	Section 5.5, 6.2	Grading Component, Grading Screen
FR-07	View grades and academic progress	Section 5.5, 6.3	Grading Component, Student Dashboard

8. APPENDICES

This section contains supporting information that assists in the interpretation and understanding of the Software Design Document. The material included in this section is referenced throughout the SDD and is consistent with the terminology and definitions used in the SRS. The appendices do not introduce any new functional or non-functional requirements. Instead, they provide clarification of terms, abbreviations, and reference documents used during system design. This section ensures consistency and avoids ambiguity in the interpretation of design elements. The information provided here supports system implementation, review, and validation activities.

8.1 Glossary

Term	Definition
Learning Management System (LMS)	A software system designed to manage courses, users, learning content, assignments, and assessments.

Software Requirements Specification (SRS)	A document that defines the functional and non-functional requirements of the system.
Software Design Document (SDD)	A document that describes the system architecture, components, interfaces, and design decisions.
Administrator	A user role responsible for managing users and courses.
Teacher	A user role responsible for course content, assignments, and grading.
Student	A user role enrolled in courses and authorized to view content and submit assignments.

8.2 Acronyms and Abbreviations

Acronym	Meaning
UI	User Interface
RTM	Requirements Traceability Matrix
UML	Unified Modeling Language

DBMS	Database Management System
CRUD	Create, Read, Update, Delete

8.3 References

1. IEEE Std 1016 – Software Design Description
2. Learning Management System – Software Requirements Specification
3. Unified Modeling Language (UML) Specification